PENETRATION TESTING REPORT

FOOTPRINTING & RECONNAISSANCE WITH MULTIPLE KALI LINUX TOOLS

W2-PM1 | WEEK 2 | NETWORKWALKS

Pentester Name	**Shaik Mahboob Mohiuddin**
Program / Batch	Cybersecurity & Ethical Hacking – Networkwalks
Date of Testing	10–11 November 2026
Module Completed	W2-PM1 – Footprinting & Reconnaissance (Multiple Kali Tools)
Client / Target	networkwalks.com (training target provided by the Networkwalks program)
Permission Secured from Client?	Yes – authorised as part of the assigned lab
Operating System	Kali Linux (running in Oracle VirtualBox)
Tools Used	WHOIS, WhatWeb, Nslookup, Curl, Wafw00f, DNSRecon
Phases Covered	**Phase 1:** Reconnaissance & Footprinting

1. Liability Disclaimer

I performed these activities only on the system I was authorised to assess as part of the Networkwalks cybersecurity program (networkwalks.com), and only for education and research purposes. Nothing in this report should be used to break the law. The instructor, the authors and Networkwalks are not responsible for how this knowledge is used; every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of employment and a permanent record. In most countries, unauthorised access is a crime even when nothing is damaged.

The work documented here consisted of information-gathering checks only. No exploitation, password testing, denial-of-service activity or vulnerability validation was performed. The observations are therefore not proof of exploitable vulnerabilities.

2. Introduction

Footprinting, also known as reconnaissance, is the information-gathering phase of an ethical hacking assessment. It allows a security professional to build an initial profile of a target from public domain-registration data, DNS records, web technologies, HTTP headers and web application firewall (WAF) information before any deeper testing begins.

This report covers Week 2, Project Module 1 (W2-PM1): Footprinting & Reconnaissance with Multiple Kali Linux Tools. I used six tools against networkwalks.com, each providing a different type of information: WHOIS (registration data), WhatWeb (web technology fingerprinting), Nslookup (DNS resolution), Curl (HTTP response headers), Wafw00f (WAF detection) and DNSRecon (DNS record enumeration).

All commands were run from a Kali Linux virtual machine. Every step below lists the exact command used, the results I observed, and a short note on why the finding matters from a security perspective. Screenshots of each command are included in Section 8 as evidence. All findings are taken from my own command output.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux	Operating system used to run all reconnaissance commands.
WHOIS	Find public domain registration details: registrar, dates, status and name servers.
WhatWeb	Fingerprint web technologies such as the web server, CMS, plugins, frameworks and IP address.
Nslookup	Resolve the domain name to its IP address using DNS.
Curl -I	Read the HTTP response headers without downloading the page body.
Wafw00f	Detect whether a Web Application Firewall is present in front of the website.
DNSRecon	Enumerate DNS records such as SOA, NS, MX, A, TXT and SRV.

4. Activities Performed

4.1 WHOIS – Domain Registration Information

Objective: Query the public registration record to identify the registrar, key dates, domain status and name servers.

Command: `whois networkwalks.com`

Observed findings:

- Registrar: GoDaddy.com, LLC (IANA ID 146).
- Created on 6 November 2019; last updated on 12 November 2025; registry expiry on 6 November 2027.
- Domain status: clientTransferProhibited, clientUpdateProhibited, clientRenewProhibited and clientDeleteProhibited (registrar lock enabled).
- Name servers: NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM.
- DNSSEC: unsigned.
- Registrant and technical contact details are privacy-protected through Domains By Proxy, LLC.

Security relevance: WHOIS data gives an initial picture of who administers the domain and where it is hosted. The registrar lock and registrant privacy protection are good practice and limit direct exposure of personal information, while the unsigned DNSSEC status is noted for later review.

Evidence: Evidence 1 and Evidence 2.

4.2 WhatWeb – Website Technology Fingerprinting

Objective: Identify the technologies exposed by the website, including the web server, CMS, plugins, frameworks and IP address.

Command: `whatweb networkwalks.com`

Observed findings:

- HTTP returned 301 Moved Permanently and redirected to HTTPS; the HTTPS site returned 200 OK.
- Web server: Apache. Page title: “Networkwalks Academy”. Country: United States.
- CMS: WordPress 7.1.4 with the WordPress Download Manager plugin, version 3.3.76 (reported in the page generator tags).
- Other technologies: jQuery 3.7.1, Google Tag Manager, HTML5 and Open Graph protocol.
- WhatWeb also reported Bootstrap 7.1.4. This value matches the WordPress version and does not correspond to a real Bootstrap release, so it is most likely a detection artefact (version taken from asset query strings) and should be verified manually before being relied upon.
- IP address: 192.232.216.135. The contact e-mail address info@networkwalks.com was also exposed.
- Cookie __wpdm_client was set with the HttpOnly flag.

Security relevance: Technology and version information helps defenders identify components that need patch review. A version string alone does not prove that a component is vulnerable.

Evidence: Evidence 3.

4.3 Nslookup – DNS Resolution

Objective: Resolve networkwalks.com to its IP address using the configured DNS resolver.

Command: `nslookup networkwalks.com`

Observed findings: The query was answered by the resolver 8.8.8.8 (port 53) as a non-authoritative answer. The domain networkwalks.com resolved to 192.232.216.135.

Security relevance: DNS resolution shows the address associated with the domain at the time of the query. It supports asset inventory and authorised assessment scoping; an IP address on its own is not a security weakness.

Evidence: Evidence 4.

4.4 Curl – HTTP Response Headers

Objective: Read the HTTP response headers to review the server response, technology disclosure, cookies, security-related headers and exposed links.

Command: `curl -I https://networkwalks.com`

Observed findings:

- Status: HTTP/2 200. The server header reported Apache.
- The link header exposed the WordPress REST API root (/wp-json/) and a page-specific route (/wp-json/wp/v2/pages/53).
- Cookie __wpdm_client was set with the Secure and HttpOnly flags.

- Present: permissions-policy and referrer-policy (no-referrer-when-downgrade). Caching headers: x-endurance-cache-level and x-nginx-cache.
- Not observed in the response: Strict-Transport-Security (HSTS), Content-Security-Policy, X-Frame-Options and X-Content-Type-Options.

Security relevance: HTTP headers can disclose implementation details and API routes. The absence of common hardening headers means the site is not instructing browsers to apply those protections; this should be reviewed against application requirements.

Evidence: Evidence 5.

4.5 Wafw00f – Web Application Firewall Detection

Objective: Determine whether a Web Application Firewall is present in front of the target website.

Command: `wafw00f networkwalks.com`

Observed findings: Wafw00f version 2.4.2 reported that https://networkwalks.com is behind a ModSecurity (SpiderLabs) WAF. The detection used 2 requests.

Security relevance: Detecting a WAF shows that a filtering layer is in place. It does not prove that all attacks are blocked or that the WAF is correctly configured.

Evidence: Evidence 6.

4.6 DNSRecon – DNS Record Enumeration

Objective: Enumerate DNS records for the domain, including SOA, NS, MX, A, TXT and SRV records.

Command: `dnsrecon -d networkwalks.com`

Observed findings:

- DNSSEC query: no answer was returned, which is consistent with the “unsigned” status in the WHOIS record.
- SOA: ns6135.hostgator.com (50.87.144.87).
- NS: ns6135.hostgator.com (50.87.144.87) and ns6136.hostgator.com (192.232.216.131).
- Both name servers disclosed their BIND software version as 9.16.23-RH.
- MX: mail.networkwalks.com (192.232.216.135). A: networkwalks.com (192.232.216.135).
- TXT: an SPF policy (v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all) and a Google site-verification record.
- SRV: eight _autodiscover._tcp records pointing to cpanelemaildiscovery.cpanel.net on port 443. DNSRecon reported “8 Records Found” for this SRV enumeration.

Security relevance: DNS records reveal parts of the domain’s hosting and e-mail configuration. Reviewing them helps confirm that published services are expected and that DNS and e-mail policies are maintained.

Evidence: Evidence 7.

5. Risk Analysis / Impact

Based on the information collected, I identified the following potential information-exposure risks. These are preliminary qualitative assessments, not confirmed vulnerabilities or formal CVSS ratings.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology and version information exposed	WhatWeb identified Apache, WordPress 7.1.4, WordPress Download Manager 3.3.76 and jQuery 3.7.1.	Exposed component and version details may help identify software that needs patch and configuration review.	● **Medium**
2	Domain registration and hosting details available	WHOIS showed GoDaddy as registrar and HostGator name servers; registrant privacy protection and registrar lock were present.	Provides administrative and hosting context for reconnaissance. Privacy protection limits personal data exposure.	● **Low**
3	Server IP address identifiable	Nslookup and WhatWeb returned 192.232.216.135.	Identifies the network location of the web service for authorised asset verification.	● **Low**
4	HTTP technical information exposed	Curl returned the Apache server header, caching headers, a cookie name and WordPress REST API links.	May assist technology fingerprinting and endpoint discovery.	● **Low**
5	HTTP security headers not observed	Curl output did not include HSTS, CSP, X-Frame-Options or X-Content-Type-Options.	Browser-side protections that these headers provide are not being requested by the server.	● **Low**
6	WAF technology identifiable	Wafw00f detected ModSecurity (SpiderLabs).	Reveals a component of the web application's security architecture.	● **Low**
7	DNS and mail infrastructure details exposed	DNSRecon returned SOA, NS, MX, A, TXT and SRV records, an SPF policy and the BIND version (9.16.23-RH) of both name servers.	Helps build a broader hosting and mail-infrastructure profile; disclosed DNS software versions can assist targeting.	● **Medium**
8	DNSSEC not enabled	WHOIS showed "DNSSEC: unsigned"; DNSRecon received no answer to its DNSSEC query.	DNS responses for the domain cannot be cryptographically validated.	● **Low**

Risk level key: ● Medium ● Low

The practical exercises involved information gathering only. No exploitation or vulnerability validation was performed. The presence of a software version, IP address, DNS record or WAF does not by itself mean that the system is vulnerable; further authorised security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on these observations, I recommend the following security improvements:

1. **Review publicly exposed technology information**
   Maintain an inventory of the CMS, plugins, frameworks and server software, and disclose only what is operationally necessary.
2. **Keep software updated**
   Check WordPress, WordPress Download Manager, jQuery and the web server against vendor security advisories and apply supported updates after testing. Verify the Bootstrap version manually, as the WhatWeb value is unreliable.
3. **Harden HTTP response headers**
   Add Strict-Transport-Security, Content-Security-Policy, X-Frame-Options (or frame-ancestors) and X-Content-Type-Options, and review the referrer policy. Confirm that exposed REST API routes are intentional.
4. **Limit DNS software disclosure**
   Ask the hosting provider to hide or restrict BIND version information on the name servers.
5. **Review DNS records regularly**
   Remove obsolete records and verify that A, MX, NS, TXT and SRV records point only to approved services.
6. **Consider enabling DNSSEC**
   Enable DNSSEC with the registrar and DNS host, where supported, to protect the integrity of DNS responses.
7. **Strengthen e-mail authentication**
   Review the SPF policy (currently soft-fail, ~all) and confirm that DKIM and DMARC are configured and monitored.
8. **Maintain and monitor the WAF**
   Confirm that ModSecurity is enabled, updated, correctly tuned and monitored; detection alone does not demonstrate effectiveness.
9. **Protect registrar and administrative accounts**
   Keep the registrar lock enabled and secure the registrar account with a strong password and multi-factor authentication.
10. **Perform follow-up testing only with authorisation**
    Any deeper scanning or vulnerability validation must be separately authorised and documented.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking program, I completed a footprinting and reconnaissance exercise against networkwalks.com using six Kali Linux tools. WHOIS provided registration and name-server information, WhatWeb identified the web server and technologies, Nslookup resolved the domain to 192.232.216.135, Curl returned the HTTP headers and WordPress REST API links, Wafw00f detected ModSecurity (SpiderLabs), and DNSRecon enumerated DNS records covering hosting, mail and service discovery.

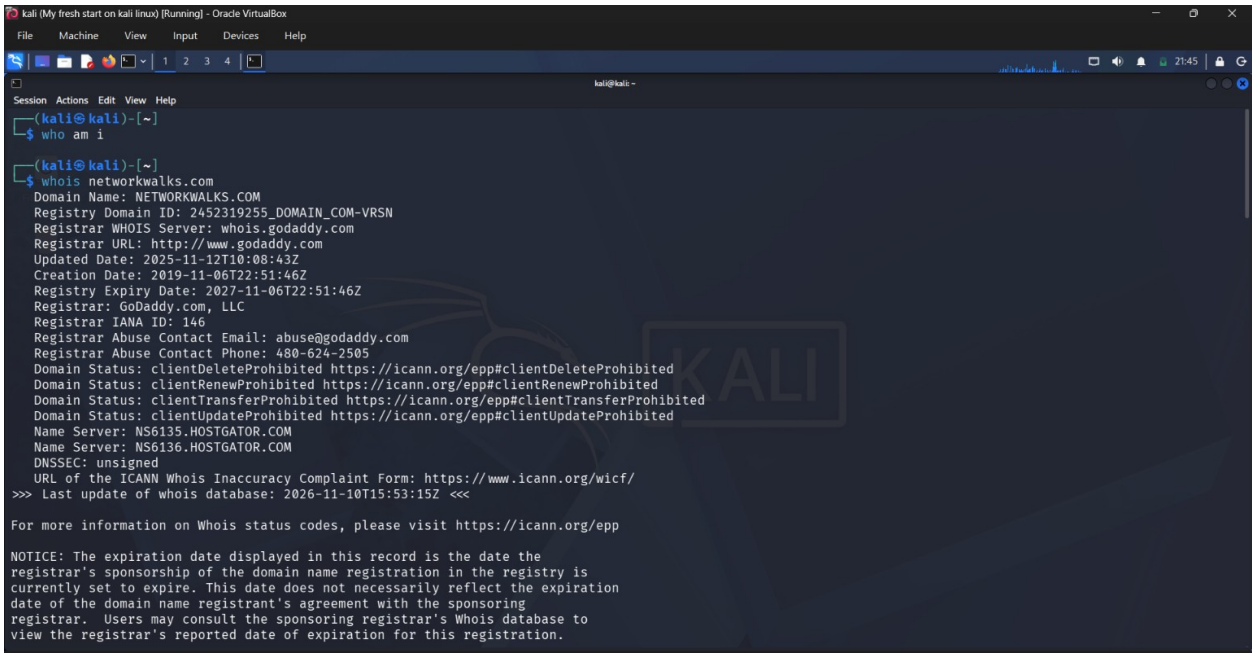
The exercise showed how each tool contributes a separate piece of information to an initial target profile, and that results from different tools should be cross-checked, as with the unreliable Bootstrap version reported by WhatWeb and the DNSSEC status confirmed by both WHOIS and DNSRecon. It also reinforced the importance of recording commands and outputs accurately and of explaining what each observation means and what risk it may create.

The results can guide a defensive review, but they do not establish that the target is vulnerable. Reconnaissance must always stay within an authorised scope, and this work was completed as part of an assigned educational cybersecurity lab.

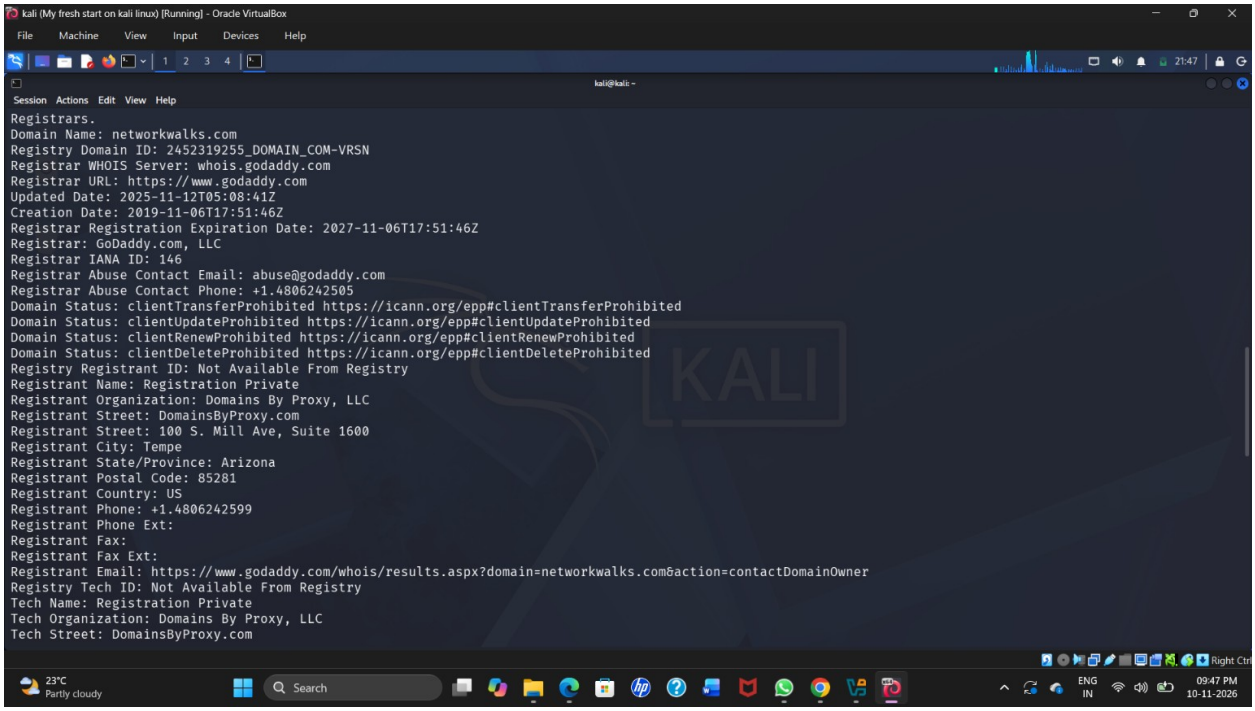
8. Evidences Collected

The following screenshots were captured in Kali Linux and show the commands and results described in Section 4.

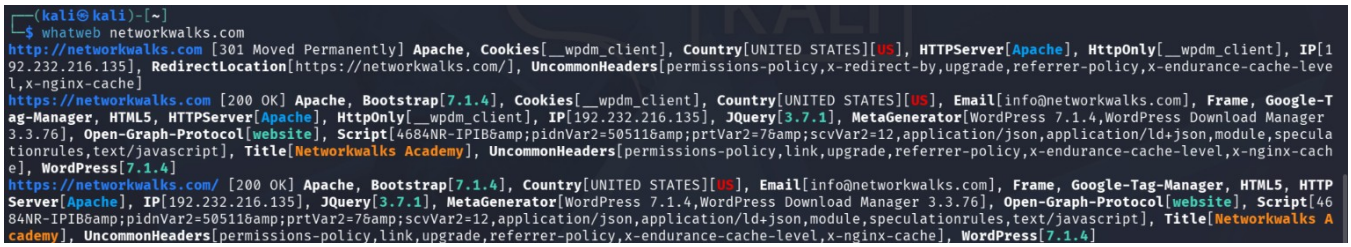
Evidence 1 – WHOIS command (registry record)



Evidence 2 – WHOIS registrant and technical details



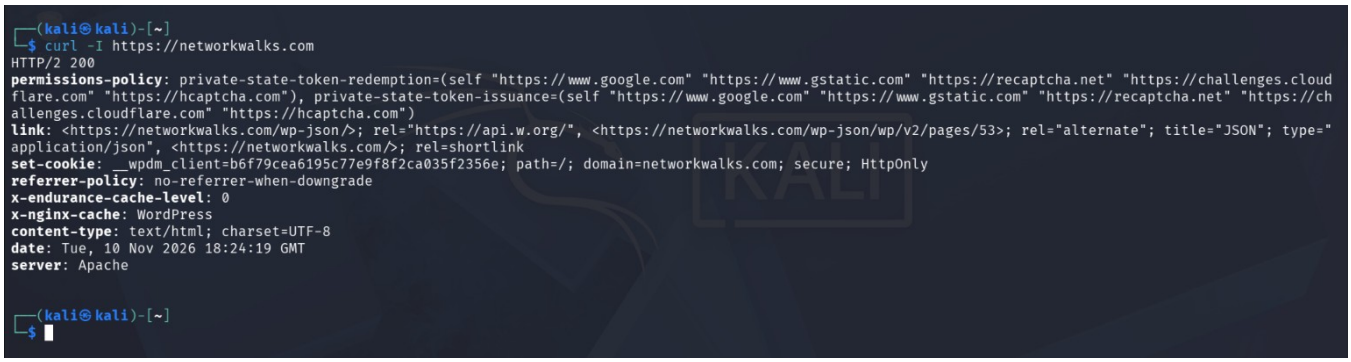
Evidence 3 – WhatWeb technology fingerprinting



Evidence 4 – Nslookup DNS resolution



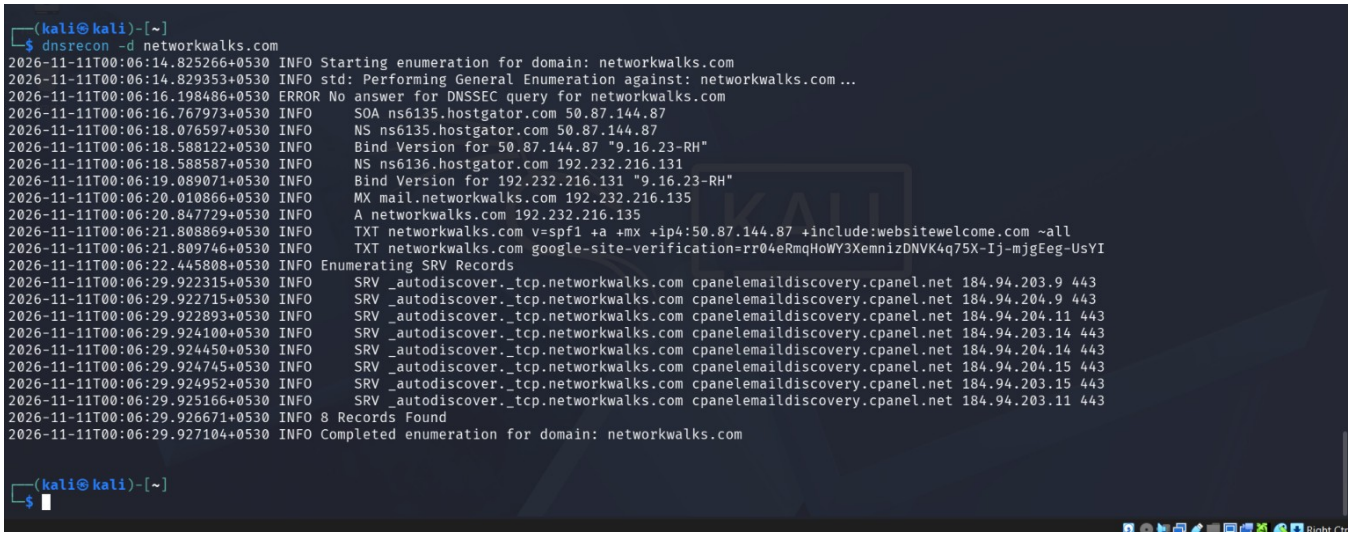
Evidence 5 – Curl HTTP response headers



Evidence 6 – Wafw00f WAF detection



Evidence 7 – DNSRecon DNS enumeration



— END OF REPORT —

Project Information

Author	Shaik Mahboob Mohiuddin
Program Name	Cybersecurity & Ethical Hacking – Networkwalks
Week / Module	Week 2 / W2-PM1
Project Title	Footprinting & Reconnaissance with Multiple Kali Linux Tools
Target	networkwalks.com
Testing Period	10–11 November 2026